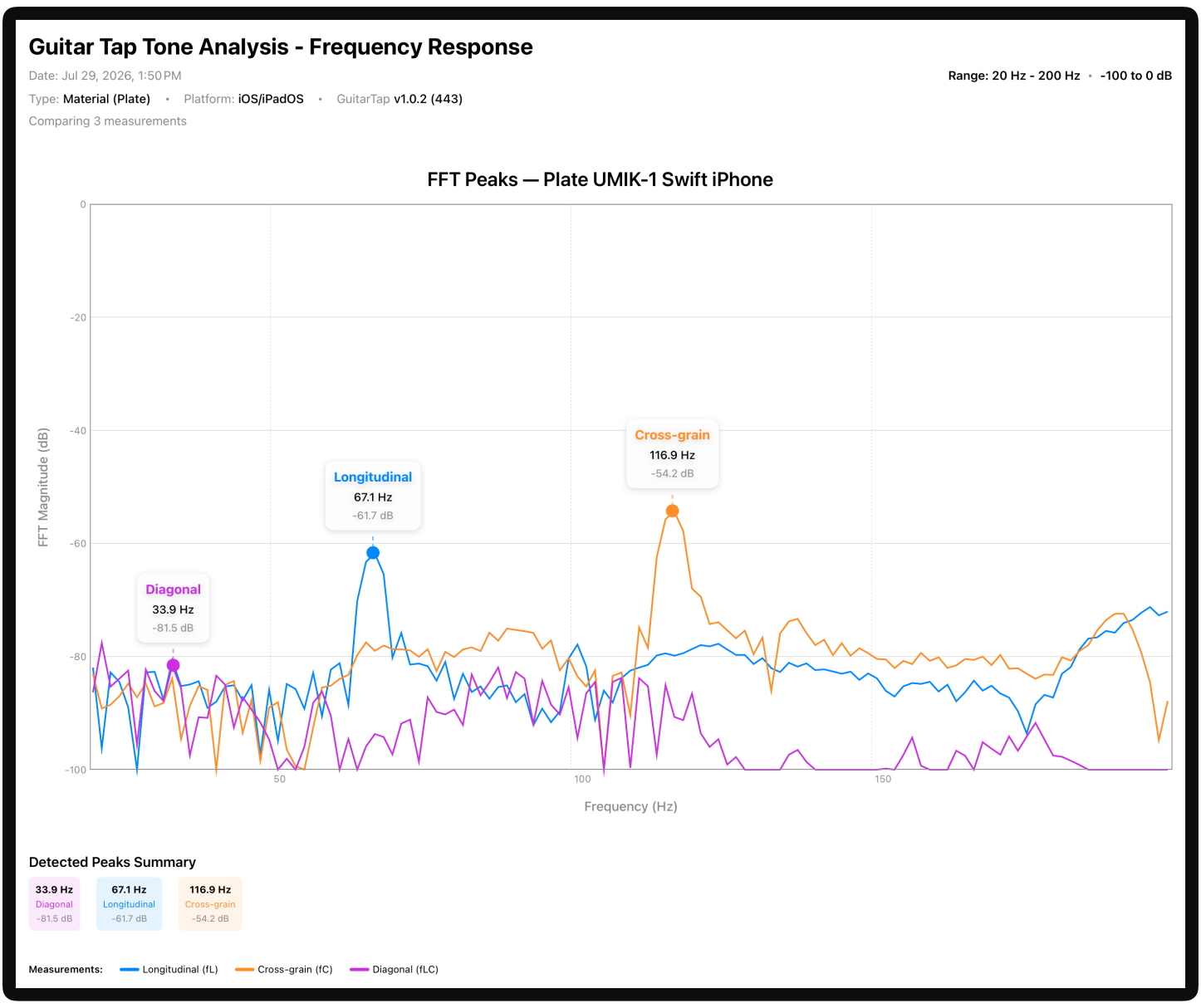
GuitarTap

Tap Tone Analysis Report

Jul 29, 2026, 1:50 PM

Measurement Name:	Plate UMIK-1 Swift iPhone
Type:	Material (Plate)
Frequency Range:	20.0 Hz – 200.0 Hz
Microphone:	Umik-1 Gain: 18dB · calibrated (7108913)

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
33.9 Hz	-81.5 dB	C#1	11.5	Diagonal (fLC)
67.1 Hz	-61.7 dB	C2	15.3	Longitudinal (fL)
116.9 Hz	-54.2 dB	A#2	26.7	Cross-grain (fC)

Sample Dimensions

Length: **557.50 mm** Width: **220.50 mm** Thickness: **4.85 mm**

Mass: **208.0 g** Density: **0.349 g/cm³**

Body Dimensions

Body Length (a): **490 mm** Lower Bout Width (b): **368 mm**

Panel Stiffness: **f_vs = 60 (Classical Top)**

Gore Target Thickness

2.82 mm

Plate Properties

Speed of Sound (L): **4182 m/s**

Speed of Sound (C): **1140 m/s**

Young's Modulus (L): **6.10 GPa**

Young's Modulus (C): **0.45 GPa**

GLC (Shear Modulus): **0.312 GPa**

Cross/Long Ratio: **0.074**
typical: 0.04–0.08

Specific Modulus (L): **17.5** (Fair)

Specific Modulus (C): **1.3** (Very Good)

Radiation Ratio (L): **12.0**

Radiation Ratio (C): **3.3**

Long/Cross Ratio: **13.5**
typical: 12–25

Overall Quality: Good

Three-Tap Measurement Process:

- **1. Longitudinal (fL) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (fC) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. Diagonal (fLC) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.